WON-YOUNG CHOI

Ph.D. Candidate in Materials Science and Engineering · Spintronics Researcher

Seoul National University | Korea Institute of Science and Technology (KIST)

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WEBSITE

RESEARCH PROFILE

Ph.D. candidate specializing in spin transport and spin-orbit torque phenomena in magnetic thin-film heterostructures. Experienced in magnetic material and interface engineering, thin-film fabrication, harmonic torque characterization, and pulse-induced magnetization switching. Current research focuses on controlling spin-current generation and symmetry using antiferromagnetic magnons for energy-efficient spintronic devices.

Research Interests: Spintronics · Spin-Orbit Torque · Magnonic Spin Transport · Antiferromagnetic Materials · Magnetic Memory Devices

RESEARCH HIGHLIGHTS

Magnonic Spin Torque

Demonstrated magnetization switching driven by magnonic spin dissipation in an antiferromagnet/ferromagnet heterostructure.

Spin-Symmetry Control

Investigating antiferromagnetic magnon spin filtering to tune spin-current polarization and spin-orbit-torque symmetry.

Materials-to-Device Research

Integrated magnetic multilayer engineering, microfabrication, electrical transport, and pulse-switching characterization.

EDUCATION

Mar. 2023 – Present **Ph.D. Candidate, Materials Science and Engineering** | Seoul National University, Seoul, Korea

- Advisor: Prof. Hyejin Jang; Co-advisor: Dr. Dong-Soo Han (KIST)

Mar. 2019 – Feb. 2021 **M.S. in Physics** | Sogang University, Seoul, Korea

- GPA: 3.76/4.3; Academic Excellence Scholarship; Advisor: Prof. Myung-Hwa Jung
- Master's thesis requirement fulfilled by a peer-reviewed SCI(E) publication.

Mar. 2012 – Feb. 2019 **B.S. in Physics, Summa Cum Laude** | Sogang University, Seoul, Korea

- GPA: 3.78/4.3; Major GPA: 4.09/4.3; ranked 3rd of 24 graduates.
- Dean's List and multiple Honors Scholarships.

RESEARCH EXPERIENCE

Mar. 2023 – Present **Student Researcher** | Center for Semiconductor Technology, KIST

- Investigate spin-current generation, transport, and magnetization switching in antiferromagnet/ferromagnet heterostructures.
- Design and fabricate magnetic multilayers and microdevices using magnetron sputtering, photolithography, and Ar-ion milling.
- Characterize spin-orbit torques using second-harmonic electrical measurements and pulse-induced switching experiments.
- Demonstrated magnetization switching driven by magnonic spin dissipation, resulting in a co-first-author publication in Nature Communications.

Oct. 2021 – Feb. 2023 **Research Intern** | Center for Spintronics, KIST

- Studied spin transport and magnetoresistance in ferromagnetic-metal/antiferromagnetic-insulator heterostructures.
- Performed thin-film fabrication and electrical transport measurements for spin Hall and unidirectional magnetoresistance studies.

Mar. 2021 – Aug. 2021 **Visiting Researcher** | Kläui Lab, Johannes Gutenberg University Mainz, Germany

- Participated in collaborative research on electrical-current control of interlayer Dzyaloshinskii–Moriya interaction in magnetic multilayers.
- Contributed to an international collaboration resulting in a Nano Letters publication.

Mar. 2018 – Feb. 2019 **Undergraduate Student Researcher** | Sogang University

- Fabricated magnetic thin films using magnetron co-sputtering and evaluated magnetic properties using SQUID-VSM/MPMS.
- Investigated electrical transport properties of magnetic thin films.

ADDITIONAL EXPERIENCE

Mar. 2019 – Jun. 2020 **Teaching Assistant** | Sogang University

- Experimental Physics I–II and Mathematical Physics I.

Oct. 2013 – Oct. 2015 **Air Materials Supply Sergeant** | Republic of Korea Air Force

- Completed military service; served as chief soldier from Jan. to Jul. 2015.

MANUSCRIPT IN PREPARATION († co-first author)

1. J.-H. Ha†, **W.-Y. Choi**†, et al., "Symmetry-Tunable Spin-Orbit Torque via Antiferromagnetic Magnon Spin Filtering." In preparation.
2. **W.-Y. Choi**†, J.-H. Ha†, et al., "Unidirectional Magnetoresistance in antiferromagnet/ferromagnet heterostructure." In preparation.

SELECTED PUBLICATIONS († co-first author, * corresponding author)

1. **W.-Y. Choi**†, J.-H. Ha†, M.-S. Jung, S. B. Kim, H.-C. Koo, O. Lee, B.-C. Min, H. Jang, A. Shahee, J.-W. Kim, M. Kläui, J.-I. Hong*, K.-W. Kim*, D.-S. Han*, "Magnetization switching driven by magnonic spin dissipation." Nature Communications 16, 5859 (2025).
2. **W.-Y. Choi**, W. Yoo, M.-H. Jung*, "Emergence of the topological Hall effect in a tetragonal compensated ferrimagnet $Mn_{2.3}Pd_{0.7}Ga$." NPG Asia Materials 13, 79 (2021).
3. **W.-Y. Choi**†, J. H. Jeon†, H. W. Bang, W. Yoo, S. K. Jerng, S. H. Chun*, S. Lee*, M.-H. Jung*, "Proximity-Induced Magnetism Enhancement Emerged in Chiral Magnet MnSi/Topological Insulator Bi_2Se_3 Bilayer." Advanced Quantum Technologies 4, 2000124 (2021).
4. **W.-Y. Choi**, H. W. Bang, S. H. Chun, S. Lee, M.-H. Jung*, "Skyrmion Phase in MnSi Thin Films Grown on Sapphire by a Conventional Sputtering." Nanoscale Research Letters 16, 7 (2021).

PRESENTATIONS

- Invited Talk — "Self-generated Spin-Orbit Torques Driven by Magnonic Spin Dissipation in Antiferromagnet/Ferromagnet Bilayer." 70th Annual Conference on Magnetism and Magnetic Materials (MMM), 2025.
- Poster — "Magnonic Spin Dissipation Driven Anisotropic Spin Torque in Easy-Axis Antiferromagnet/Ferromagnet Bilayer." Korean Magnetics Society Summer Conference, 2025.
- Poster — "Spin-Orbit Torque by Magnon Dissipation." 11th International Symposium on Metallic Multilayers (MML), 2023.

- Poster — “Spin Hall Magnetoresistance and Unidirectional Magnetoresistance in Ferromagnetic-Metal/Antiferromagnetic-Insulator Bilayers.” Korean Magnetics Society Summer Conference, 2023.
- Poster — “Spin-Orbit Torque by Magnon Dissipation.” Korean Magnetics Society Winter Conference, 2022.
- Poster — “Topological Hall Effect in a Mixed Phase of Mn_3Ga Thin Film.” Korean Magnetics Society Summer Conference, 2022.
- Oral — “Observation of Unconventional Anomalous Hall Effect in Compensated Mn2.3Pd0.7Ga Thin Film.” Korean Physical Society Fall Meeting, 2020.
- Poster — “Compensated Ferrimagnetism of $Mn_{2+\delta-x}Pd_xGa$ Thin Film.” Korean Physical Society Fall Meeting, 2019.

HONORS & AWARDS

Jul. 2026	STAR Outstanding Graduate Student Award, Korean Magnetics Society
Dec. 2025	PSI Excellent Student Researcher Award, KIST Post-Silicon Semiconductor Institute
Oct. 2025	Condensed Matter Physics MSM Award, Korean Physical Society Fall Meeting
Nov. 2022	Best Poster Award, Korean Magnetics Society Winter Conference
May 2022	Poster Award, Korean Magnetics Society Summer Conference
Jan. 2021	International Academic Papers Scholarship, Ministry of Education, Republic of Korea
Oct. 2020	Outstanding Presentation Award, Korean Physical Society Fall Meeting
Mar. 2019	Academic Excellence Scholarship, Graduate School of Sogang University
Mar. 2018	CHO-LEE Scholarship
Jan. 2018	Dean’s List Award, Sogang University — Top 3%
Sep. 2016 – Dec. 2017	Honors Scholarships, Sogang University

SOFTWARE DEVELOPMENT

Figure Creator v7 (macOS, 2026): Developed a desktop application for publication-ready scientific visualization, supporting 2D, polar, 3D, heatmap, surface, bar, and multi-panel figures; interactive layout and annotation editing; LaTeX labels, error bars, broken axes, journal-size presets, and high-resolution export. Released with English and Korean interfaces.

TopDock v0.1.0 (macOS, Swift/SwiftUI/AppKit, 2026): Developed a zero-dependency menu-bar quick launcher activated from the MacBook notch or a top-center screen hotspot, featuring in-panel folder navigation, search, workspaces, Quick Look, drag-and-drop file handling, a global keyboard shortcut, and external-display support.

Macrospin Simulator (Python, in development): Developing a desktop simulator that numerically solves the Landau–Lifshitz–Gilbert equation and visualizes magnetization precession, spin-orbit-torque switching, and time-resolved trajectories within the macrospin approximation.

TECHNICAL SKILLS

Thin-Film & Device Fabrication: Magnetron sputtering, co-sputtering, photolithography, Ar-ion milling, wire bonding, Hall-bar device fabrication

Spintronic Device Characterization: Second-harmonic spin-orbit-torque measurements, pulse-induced magnetization switching, electrical transport, Van der Pauw measurements

Magnetic & Structural Characterization: MPMS/SQUID-VSM, PPMS, XRD, SEM, energy-dispersive X-ray spectroscopy

Programming & Measurement Automation: Python, C, LabVIEW

Scientific & Engineering Software: OriginLab, QtiPlot, VESTA, Elmer FEM, NanoCAD, FreeCAD, Adobe Illustrator, Blender

Languages: Korean — native; English — fluent; Japanese — intermediate

FULL PUBLICATIONS († co-first author, * corresponding author)

1. S. Jena, R. Urkude, **W.-Y. Choi**, M.-H. Jung, K. Amemiya, B. Ghosh, V. R. Singh*, "Giant strain-induced magnetic anisotropy in the skyrmionic-textured MnSi thin films via polarization-dependent X-ray magnetic circular dichroism and X-ray near-edge spectroscopy." Physical Chemistry Chemical Physics (2026). DOI: 10.1039/d6cp00805d
2. J.-B. Choi, H. Seong, **W.-Y. Choi**, S.-J. Noh, J.-S. Lee, H.-S. Lee, Y.-C. Kim, O.-J. Lee, B.-C. Min, J.-H. Kwon*, D.-S. Han*, "Spin Hall Magnetic Sensors with Dynamic Offset Calibration and Structurally Tunable Sensitivity." IEEE Sensors Journal 25(21), 39605–39611 (2025).
3. **W.-Y. Choi**†, J.-H. Ha†, M.-S. Jung, S. B. Kim, H.-C. Koo, O. Lee, B.-C. Min, H. Jang, A. Shahee, J.-W. Kim, M. Kläui, J.-I. Hong*, K.-W. Kim*, D.-S. Han*, "Magnetization switching driven by magnonic spin dissipation." Nature Communications 16, 5859 (2025).
4. S. Yu, J. Park, Y. Jo, W. Kim, H. Kim, **W.-Y. Choi**, M. Kim, D.-S. Han, M.-H. Jung, K. Rhie, K. Lee*, "The control of magnetization reversal by tuning the interlayer exchange interaction in magnetic multilayers." Journal of the Korean Physical Society 86, 62–67 (2024).
5. H. Seong, **W.-Y. Choi**, J. Choi, D.-H. Kim, T.-E. Park, B.-C. Min, H.-J. Choi, D.-S. Han*, "Magnetic properties of GdFeCo thin films tailored by sputtering conditions." Current Applied Physics 68, 131–137 (2024).
6. S. Jena, R. Urkude, **W.-Y. Choi**, K. K. Pandey, S. Karwai, M. H. Jung, J. Gardner, B. Ghosh, V. R. Singh*, "Electronic Structures of Skyrmionic Polycrystalline MnSi Thin Film Studied by Resonance Photoemission and X-ray Near Edge Spectroscopy." Journal of Applied Physics 135, 165301 (2024).
7. S. Jena*, R. Dawn, **W.-Y. Choi**, Y. Singh, A. Ghosh, S. K. Sahoo, M. H. Jung, J. Gardner, V. K. Verma, K. Amemiya, V. R. Singh*, "Texture, Electronic and Magnetic States of Mn Atoms in Polycrystalline MnSi Thin Films Facing c-Sapphire Substrate Studied by Soft X-ray Magnetic Circular Dichroism." Thin Solid Films 786, 140097 (2023).
8. F. Kammerbauer, **W.-Y. Choi**, F. Freimuth, K. Lee, F. Robert, D.-S. Han, H. J. M. Swagten, Y. Mokrousov, M. Kläui*, "Controlling the Interlayer Dzyaloshinskii–Moriya Interaction by Electrical Currents." Nano Letters 23(15), 7070–7075 (2023).
9. S. Jena, **W.-Y. Choi**, J. Gardner, M. H. Jung, S. K. Srivastava, V. K. Verma, K. Amemiya, V. R. Singh*, "Evolution of bulk magnetic structure in MnSi thin film: a soft X-ray magnetic circular dichroism study." Physica Scripta 98, 075927 (2023).
10. **W.-Y. Choi**, W. Yoo, M.-H. Jung*, "Emergence of the topological Hall effect in a tetragonal compensated ferrimagnet $Mn_{2.3}Pd_{0.7}Ga$." NPG Asia Materials 13, 79 (2021).
11. **W.-Y. Choi**†, J. H. Jeon†, H. W. Bang, W. Yoo, S. K. Jerng, S. H. Chun*, S. Lee*, M.-H. Jung*, "Proximity-Induced Magnetism Enhancement Emerged in Chiral Magnet MnSi/Topological Insulator Bi_2Se_3 Bilayer." Advanced Quantum Technologies 4, 2000124 (2021).
12. **W.-Y. Choi**, H. W. Bang, S. H. Chun, S. Lee, M.-H. Jung*, "Skyrmion Phase in MnSi Thin Films Grown on Sapphire by a Conventional Sputtering." Nanoscale Research Letters 16, 7 (2021).